

# Evaluation Design and Data Source Heterogeneity Outweigh Algorithm Choice in Ames Mutagenicity QSAR

Seungmin Yoon, Sanghun Kim, and Hyunpyo Jeon\*

*Department of Biosafety, Kyungsoong University, Busan, Republic of Korea*

E-mail: hpjeon@ks.ac.kr

## Abstract

QSAR models for Ames mutagenicity are increasingly used in regulatory assessments under EU REACH, K-REACH, and ICH M7,<sup>1-3</sup> yet reported performance varies widely with evaluation methodology. We systematically evaluated 375 model configurations—seven algorithms, five feature representations, and three preprocessing scenarios—across five genotoxicity datasets: Ames ( $n = 13,560$ ), in vitro chromosome aberration ( $n = 1,619$ ), in vivo micronucleus ( $n = 2,153$ ), and two negative-sampled variants. Using a single locked configuration (XGBoost, compact descriptors, raw SMILES) at three levels of evaluation rigor, Ames MCC fell from 0.670 (random-split CV) to 0.624 (scaffold-split CV) to 0.234 (drug→industrial leave-domain-out). The split-method effect ( $\Delta = 0.046$ ) was substantially smaller than the source-heterogeneity effect ( $\Delta = 0.390$ ): a naive source classifier reached  $\text{MCC} = 0.608$  without any structural information, reflecting a 19.5-fold prevalence gap between drug-sourced (58.3%) and industrial-sourced (3.0%) compounds.

Threshold optimization explained only part of this gap, indicating that prior shift accounts for only a subset of the LDO collapse. Algorithm robustness under source

shift was uneven: Random Forest maintained  $\text{MCC} = 0.325$  in the hardest direction where XGBoost collapsed to 0.122. In vivo micronucleus models showed ranking ability ( $\text{ROC-AUC} = 0.860$ ,  $\text{PR-AUC} = 0.237$ ) but unreliable classification ( $\text{MCC}$  95% CI crossing zero). We recommend multi-level evaluation with source-awareness as a reporting standard for Ames mutagenicity QSAR.

## Acknowledgement

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## Introduction

Under EU REACH<sup>1</sup> and the Korean Act on the Registration and Evaluation of Chemicals (K-REACH),<sup>2</sup> industrial chemical manufacturers and importers are required to assess genotoxic potential as part of their registration dossiers, and in silico QSAR predictions are explicitly accepted as supporting evidence when generated under validated models with sufficient applicability domain coverage.<sup>4</sup> For pharmaceuticals, the ICH M7(R2) guideline plays an analogous role by endorsing statistical QSAR alongside rule-based expert systems for assessing the mutagenic potential of impurities.<sup>3</sup> Across both regulatory frameworks, the Ames bacterial reverse mutation assay<sup>5,6</sup> is the central predictive target, with in vitro chromosome aberration (CA) and in vivo micronucleus (MN)<sup>7,8</sup> serving as higher-tier endpoints. Reliable models could reduce animal testing in line with the 3Rs principle.<sup>9</sup>

Published Ames QSAR models report a wide spread of metrics—accuracies of 0.80–0.90 and  $\text{MCC}$  of 0.60–0.80<sup>10–13</sup>—but these come from different splits, datasets, preprocessing pipelines, and protocols, making direct comparison difficult. At least three factors can inflate apparent performance.

**Splitting strategy.** Random splits allow structurally similar compounds (sharing the same Murcko scaffold<sup>14</sup>) to appear in both training and test, producing optimistic estimates.

Scaffold-based splitting<sup>15,16</sup> is more realistic, but the magnitude of the random-vs-scaffold gap has not been systematically quantified across genotoxicity endpoints.

**Source mixing.** Most large Ames datasets aggregate compounds from pharmaceutical libraries and industrial chemical registries,<sup>10,12</sup> which differ in composition and prevalence, and likely also in data quality and labeling conventions. When mixed, models may learn source membership rather than toxicity mechanisms—a form of source heterogeneity noted<sup>17,18</sup> but never systematically measured.

**Preprocessing.** Salt stripping, metal removal, and charge normalization can substantially alter molecular representations,<sup>19,20</sup> yet these steps are often undocumented or applied uniformly across endpoints.

Most studies evaluate a single endpoint with a single split, focus on algorithm comparison,<sup>21,22</sup> rarely test cross-domain generalization, and seldom report statistical significance.<sup>23</sup> The relative contributions of split method, source composition, and preprocessing have not been disentangled in a single consistent framework.

We address this by (i) introducing a three-level evaluation framework—scaffold-split test, random vs. scaffold CV, and cross-source leave-domain-out—applied to a single locked configuration; (ii) showing that source heterogeneity from mixing drug-like and industrial compounds (19.5-fold prevalence gap) drove a scaffold-to-LDO MCC drop of 0.390, an order of magnitude larger than the split-method effect ( $\Delta = 0.046$ ); (iii) finding that in-domain algorithm differences are small (MCC range = 0.063), but cross-source robustness varies sharply (RF 0.325 vs. XGBoost 0.122); (iv) reporting endpoint-specific preprocessing effects, with salt stripping improving the best in vivo model by +0.22 MCC while harming in vitro; and (v) releasing an open-source pipeline for full reproducibility.

# Materials and Methods

## Datasets

We evaluated five genotoxicity datasets (Table 1): three primary endpoints—Ames bacterial reverse mutation ( $n = 13,560$ ), in vitro chromosome aberration ( $n = 1,619$ ), and in vivo micronucleus ( $n = 2,153$ )—and two negative-sampled variants treated as exploratory.

The Ames dataset combines pharmaceutical (drug-like) and industrial chemical sources. Compounds carry a source label based on their originating dataset; we use “source” to emphasize that this label reflects data provenance—encompassing potential differences in chemical space, data quality, and labeling conventions—rather than a computed chemical classification. Drug-sourced compounds ( $n = 6,416$ ) had a positive rate of 58.3%, while industrial-sourced ( $n = 7,144$ ) had 3.0%, a 19.5-fold difference. The in vitro and in vivo datasets lacked source annotations.

Raw study records for all three endpoints were first mapped to endpoint categories (Ames, in vitro CA, in vivo MN) using a unified classification table spanning OECD, EPA, and EU Test Guidelines; the full guideline-to-endpoint mapping is provided in the supplementary materials. For industrial chemicals sourced from ECHA dossiers, we restricted inclusion to studies with Klimisch reliability scores of 1 (reliable without restriction) or 2 (reliable with restrictions) where available,<sup>24</sup> excluding records with score 3 or 4. When a single compound had both positive and negative study records of equal Klimisch score, we adopted a conservative hazard-assessment convention consistent with EU REACH and K-REACH guidance<sup>1,2</sup> and labeled the compound as positive. Compounds with only irreconcilable label conflicts across distinct sources were resolved by exclusion (169 Ames, 3 in vitro, 2 in vivo).

Cross-endpoint overlap analysis revealed substantial sharing: 97.5% of in vitro and 92.1% of in vivo compounds also appear in Ames, with moderate label concordance (Cohen’s  $\kappa = 0.14$ – $0.20$ ).

All datasets were split 80/20 using scaffold-based splitting<sup>15</sup> with zero scaffold-group overlap. Murcko scaffolds<sup>14</sup> were generated with RDKit;<sup>25</sup> acyclic compounds were clustered by agglomerative clustering on Morgan fingerprint<sup>26</sup> similarity.

Table 1: Dataset summary. Positive rate refers to the training partition. The 375 total configurations comprise 5 datasets  $\times$  3 preprocessing scenarios  $\times$  (6 tabular models  $\times$  4 feature modes + 1 GNN).

| Dataset                            | Assay                      | $n$    | Train  | Test  | Pos. rate (%)          | Scaffolds |
|------------------------------------|----------------------------|--------|--------|-------|------------------------|-----------|
| Ames                               | Bacterial reverse mutation | 13,560 | 10,848 | 2,712 | 29.1                   | 1,691     |
| In vitro CA                        | Chromosome aberration      | 1,619  | 1,295  | 324   | 10.4                   | 353       |
| In vivo MN                         | Micronucleus               | 2,153  | 1,722  | 431   | 3.7 / 2.8 <sup>a</sup> | 410       |
| In vitro CA (sampled) <sup>b</sup> | Neg. sampled CA            | 345    | 276    | 69    | 49.6                   | 169       |
| In vivo MN (sampled) <sup>b</sup>  | Neg. sampled MN            | 231    | 184    | 47    | 33.3                   | 129       |

<sup>a</sup>Training / test positive rate. <sup>b</sup>Supplementary exploratory datasets.

## Preprocessing, Features, and Algorithms

Each dataset was processed under three scenarios: **raw** (original SMILES), **salt-stripped** (largest fragment retained via RDKit’s `LargestFragmentChooser`<sup>25</sup> followed by charge normalization—modifying 45% of Ames, 81% of in vitro, and 83% of in vivo SMILES), and **no-metal** (compounds with metal atoms excluded).

We evaluated five feature representations. **Compact** (138 features) combined 13 physicochemical descriptors (MW, LogP, TPSA, HBD, HBA, rotatable bonds, fraction C<sub>sp3</sub>, ring counts, heteroatoms, heavy atoms, formal charge, valence electrons) with 125 structural alerts based on Benigni–Bossa SA1–SA31,<sup>27,28</sup> Kazius toxicophores,<sup>29</sup> and ISS Ames alerts. **Broad FP-512/1024/2048** (266/394/522 features) extended compact with 128/256/384 Morgan fingerprint bits<sup>26</sup> selected by mutual information<sup>30</sup> on the training set only (bits with prevalence below 1% or above 99% excluded). **Graph** representations used molecular graphs with 9-dimensional atom features and normalized adjacency matrices, paired exclusively with the GCN.

Seven classifiers were evaluated: XGBoost,<sup>31</sup> LightGBM,<sup>32</sup> Random Forest,<sup>33</sup> SVM with

RBF kernel,<sup>34</sup> Logistic Regression,<sup>35</sup> a 128–64 MLP,<sup>36</sup> and a three-layer GCN<sup>37</sup> with Layer-Norm and global mean pooling. SVM and Logistic Regression used standard scaling. Class imbalance was handled via `scale_pos_weight` or `class_weight='balanced'`. Hyperparameters were tuned via `RandomizedSearchCV` with `GroupKFold`.<sup>38</sup>

## Three-Level Evaluation Framework

For fair comparison across evaluation methods, we defined a single locked configuration—XGBoost, compact features, raw SMILES—and applied it at three levels (Figure 1):

**Level 1 (in-domain test):** the scaffold-split held-out test set, with bootstrap 95% CIs (500 iterations) on MCC,<sup>39</sup> ROC-AUC, PR-AUC, balanced accuracy, sensitivity, specificity, and Brier score.<sup>40</sup>

**Level 2 (split-method comparison):** 5-fold stratified random CV and 5-fold scaffold-grouped CV, each with 5 repeats, on the full dataset.

**Level 3 (cross-source LDO):** for Ames, trained on one source and tested on the other, using the locked configuration plus LightGBM and Random Forest to assess algorithm-specific robustness.

## Additional Analyses

Beyond the locked configuration, all 375 combinations were evaluated on the scaffold-split test set ( $5 \times 3 \times [6 \times 4 + 1] = 375$ ). Applicability domain (AD) used Tanimoto similarity on 1024-bit Morgan fingerprints<sup>26</sup> with threshold  $\geq 0.3$ . SHAP values<sup>41</sup> (`TreeExplainer`) were computed for tree-based models on the three primary endpoints. Learning curves spanned 10%–100% training fractions with 5-fold scaffold CV under the locked configuration. McNemar’s test<sup>42</sup> with continuity correction compared the top two models per endpoint.

**LDO threshold decomposition.** To disentangle prior shift from genuine source shift, we computed LDO MCC at three thresholds: *default* ( $t = 0.5$ ), *prior-corrected* ( $t =$  training positive rate, optimal under pure prior shift), and *oracle* ( $t$  maximizing MCC on the test

set, an upper bound on threshold-based recalibration). The implementation is provided as `ldo_threshold_analysis.py` in the supplementary materials.

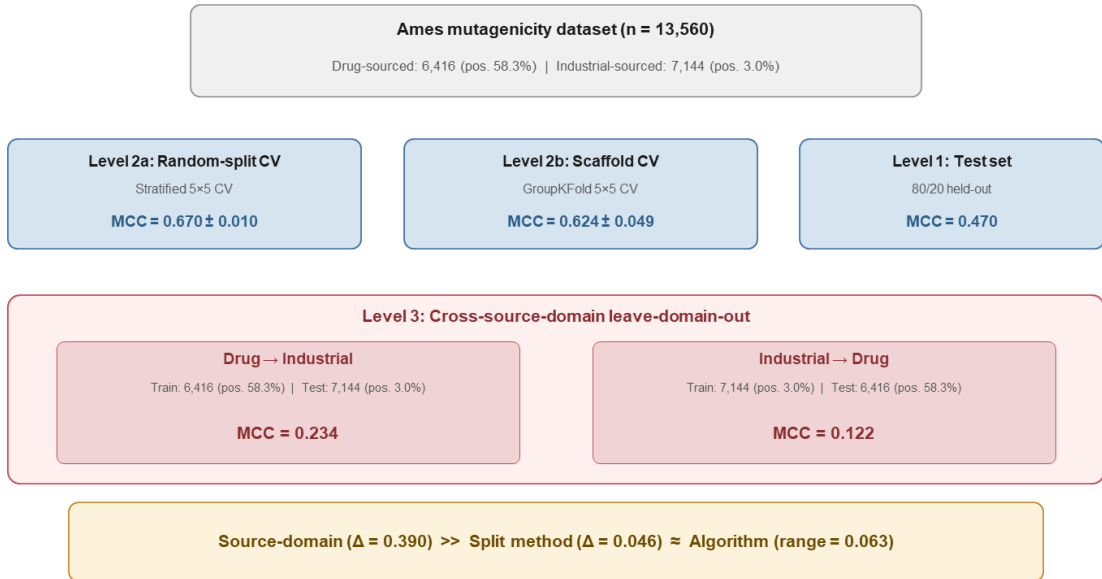


Figure 1: Three-level evaluation framework. All levels use the same locked configuration (XGBoost, compact features, raw SMILES), enabling fair MCC comparison across evaluation methods.

## Results

### Best Observed Performance Across the Full Grid

Table 2 reports the best observed performance per primary endpoint across all 375 configurations. These values are an upper bound from extensive search and should be read as a descriptive benchmark, not unbiased estimates; the confirmatory analysis uses the locked configuration (Table 4).

Under fixed features and preprocessing (compact, raw SMILES), all seven algorithms on Ames spanned an MCC range of only 0.063 (Table 3, Figure 2); the five tabular models excluding Logistic Regression spanned 0.023. Pairwise McNemar tests<sup>42</sup> between the top

Table 2: Best observed performance per primary endpoint (scaffold-split test set, full grid). Best among 375 configurations; descriptive benchmark, not unbiased estimates.

| Endpoint | Algo. | Preproc.  | Features ( $n$ ) | MCC [95% CI]       | AUC   | PR-AUC | Sens. | Spec. |
|----------|-------|-----------|------------------|--------------------|-------|--------|-------|-------|
| Ames     | XGB   | raw       | FP-2048 (522)    | 0.555 [.522,.588]  | 0.850 | 0.765  | 0.756 | 0.822 |
| In vitro | LGBM  | salt_str. | FP-1024 (394)    | 0.297 [.131,.450]  | 0.665 | 0.278  | 0.349 | 0.929 |
| In vivo  | XGB   | salt_str. | FP-512 (266)     | 0.311 [−.014,.571] | 0.860 | 0.237  | 0.319 | 0.983 |

PR-AUC baselines (prevalence): Ames 0.302, in vitro 0.124, in vivo 0.028.

two models found no significant differences (Ames  $p = 0.39$ ; in vivo  $p = 0.72$ ). This in-domain equivalence did not extend to cross-source settings (Section:Source Heterogeneity and Algorithm-Specific Robustness). The GCN underperformed fingerprint-based approaches across endpoints (Ames mean MCC 0.402 vs. 0.507 for XGBoost), consistent with prior reports that graph networks need substantially larger datasets to outperform engineered features.

Table 3: All-algorithm comparison (compact features, raw SMILES, Ames scaffold-split test). Range across all 7 algorithms: 0.063; range excluding LR and GCN: 0.023.

| Algorithm           | MCC   | ROC-AUC |
|---------------------|-------|---------|
| Random Forest       | 0.492 | 0.838   |
| ANN (MLP)           | 0.489 | 0.834   |
| LightGBM            | 0.479 | 0.833   |
| SVM (RBF)           | 0.470 | 0.832   |
| XGBoost             | 0.470 | 0.825   |
| GCN                 | 0.445 | 0.815   |
| Logistic Regression | 0.430 | 0.800   |

## Evaluation Rigor Cascade

For Ames, random-split CV reached  $\text{MCC} = 0.670 \pm 0.010$ , consistent with published benchmarks,<sup>10,13</sup> while scaffold-split CV dropped to  $0.624 \pm 0.049$  ( $\Delta = 0.046$ ) and the held-out scaffold-split test to 0.470 (Table 4, Figure 3). Cross-source LDO produced a much larger collapse: drug  $\rightarrow$  industrial reached 0.234, and industrial  $\rightarrow$  drug only 0.122. The source-heterogeneity gap ( $\Delta = 0.390$ ) was an order of magnitude larger than the split-method gap

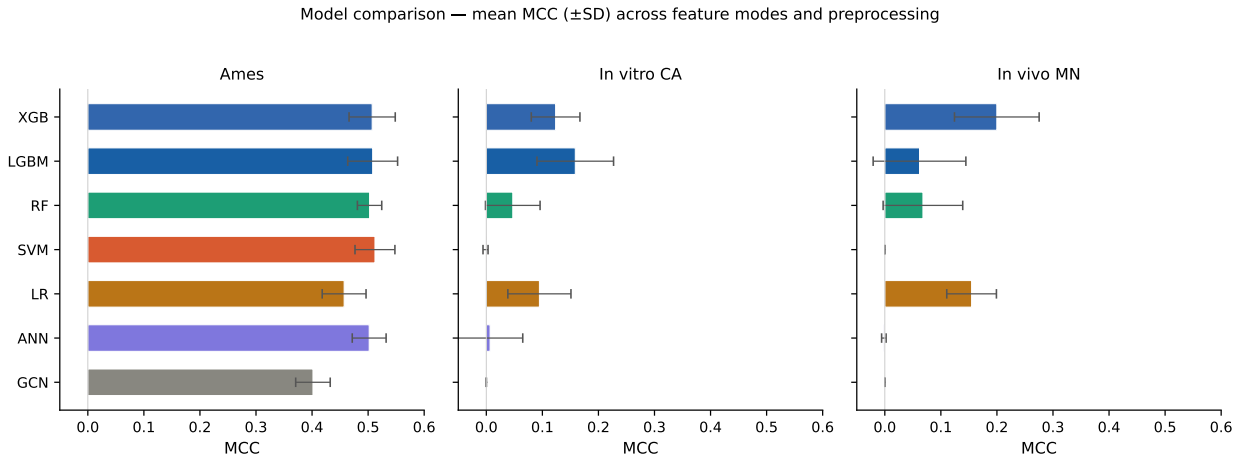


Figure 2: Mean MCC ( $\pm$ SD) by algorithm across all feature modes and preprocessing scenarios, grouped by endpoint.

( $\Delta = 0.046$ ). The LDO comparison involves additional structural factors—directional asymmetry, different test sizes, and different class distributions—so  $\Delta = 0.390$  is best read as a combined effect, but the magnitude difference between the two effects is unambiguous. For in vitro CA, random and scaffold splits gave nearly identical results (0.180 vs. 0.189); for in vivo MN, both values were near random given their standard deviations.

Table 4: Evaluation rigor cascade (locked configuration: XGBoost, compact features, raw SMILES). Level 2: mean  $\pm$  SD (25 folds). Level 1: MCC [95% bootstrap CI]. Level 3: Ames only.

| Level | Evaluation method                  | Ames MCC          | In vitro MCC       | In vivo MCC        |
|-------|------------------------------------|-------------------|--------------------|--------------------|
| 2a    | Random-split 5 $\times$ 5 CV       | 0.670 $\pm$ 0.010 | 0.180 $\pm$ 0.053  | 0.181 $\pm$ 0.094  |
| 2b    | Scaffold-split 5 $\times$ 5 CV     | 0.624 $\pm$ 0.049 | 0.189 $\pm$ 0.084  | 0.104 $\pm$ 0.111  |
| 1     | Scaffold-split test                | 0.470 [.434,.502] | 0.114 [−.023,.270] | 0.071 [−.029,.285] |
| 3a    | LDO: drug $\rightarrow$ industrial | 0.234             | —                  | —                  |
| 3b    | LDO: industrial $\rightarrow$ drug | 0.122             | —                  | —                  |

## Source Heterogeneity and Algorithm-Specific Robustness

Drug-sourced compounds ( $n = 6,416$ , positive rate 58.3%) and industrial-sourced compounds ( $n = 7,144$ , positive rate 3.0%) differed by a factor of 19.5 in mutagenicity prevalence (Figure 4). A naive classifier predicting positive for drug-sourced and negative for industrial-

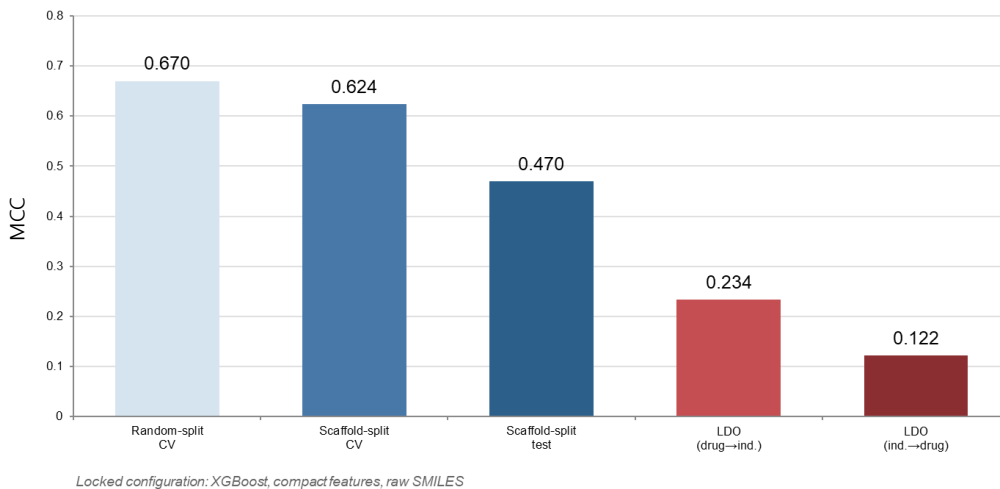


Figure 3: Evaluation rigor cascade for Ames mutagenicity (locked configuration). The source-heterogeneity effect ( $\Delta = 0.390$ ) substantially exceeds the split-method effect ( $\Delta = 0.046$ ).

sourced compounds achieved  $\text{MCC} = 0.608$  and balanced accuracy = 0.834 without any structural information—comparable to the random-split CV MCC of 0.670.

LDO with three algorithms (Table 5) revealed two patterns. Drug  $\rightarrow$  industrial generalization was uniform across algorithms (MCC 0.232–0.235). In contrast, industrial  $\rightarrow$  drug exposed sharp differences: XGBoost reached only  $\text{MCC} = 0.122$  (sensitivity 0.062), LightGBM 0.134, while Random Forest maintained  $\text{MCC} = 0.325$  (sensitivity 0.466). We hypothesize that Random Forest’s better cross-source robustness reflects feature bagging,<sup>33</sup> which reduces reliance on source-correlated feature subsets.

Table 5: Leave-domain-out results by algorithm (Ames). Drug:  $n = 6,416$ , pos. rate 58.3%. Industrial:  $n = 7,144$ , pos. rate 3.0%. Naive source classifier:  $\text{MCC} = 0.608$ .

| Algorithm     | Drug $\rightarrow$ Ind. MCC | Ind. $\rightarrow$ Drug MCC | Ind. $\rightarrow$ Drug Sens. |
|---------------|-----------------------------|-----------------------------|-------------------------------|
| XGBoost       | 0.234                       | 0.122                       | 0.062                         |
| LightGBM      | 0.232                       | 0.134                       | 0.081                         |
| Random Forest | 0.235                       | 0.325                       | 0.466                         |

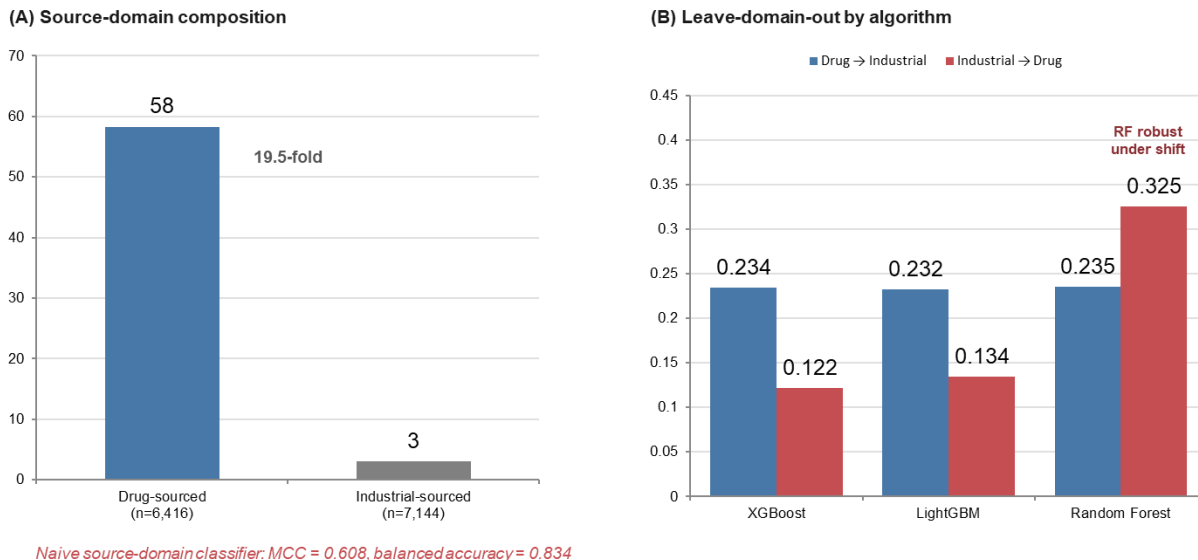


Figure 4: Source heterogeneity in Ames. (A) Positive rate by source: drug 58.3% vs. industrial 3.0% (19.5-fold). (B) LDO MCC by algorithm; Random Forest is substantially more robust in the industrial→drug direction.

## Preprocessing, In Vivo Behavior, and Feature Importance

**Preprocessing.** Salt stripping was strongly endpoint-specific (Figure 5). For in vivo MN, the best salt-stripped model reached  $MCC = 0.311$  versus 0.086 for the best raw model (+0.225); salt fragments were present in 83% of in vivo SMILES, and among salt-containing compounds the positive rate was 2.8% versus 7.1% for non-salt compounds, suggesting salts were disproportionately attached to non-mutagenic compounds. For Ames, salt stripping had mixed effects ( $\Delta = -0.03$  to  $+0.02$ ); for in vitro CA it was generally detrimental (mean  $MCC$   $0.081 \rightarrow 0.056$ ). This argues against uniform preprocessing pipelines<sup>19</sup> and in favor of per-endpoint optimization.

**In vivo micronucleus.** The best in vivo model achieved  $ROC-AUC = 0.860$  and  $PR-AUC = 0.237$  (an 8.5-fold enrichment over the 0.028 baseline), but  $MCC = 0.311$  with 95% CI crossing zero ( $-0.014$  to  $0.571$ ) indicates that classification at any fixed threshold is unreliable. The test set contains only 12 positives among 431 compounds. The locked-

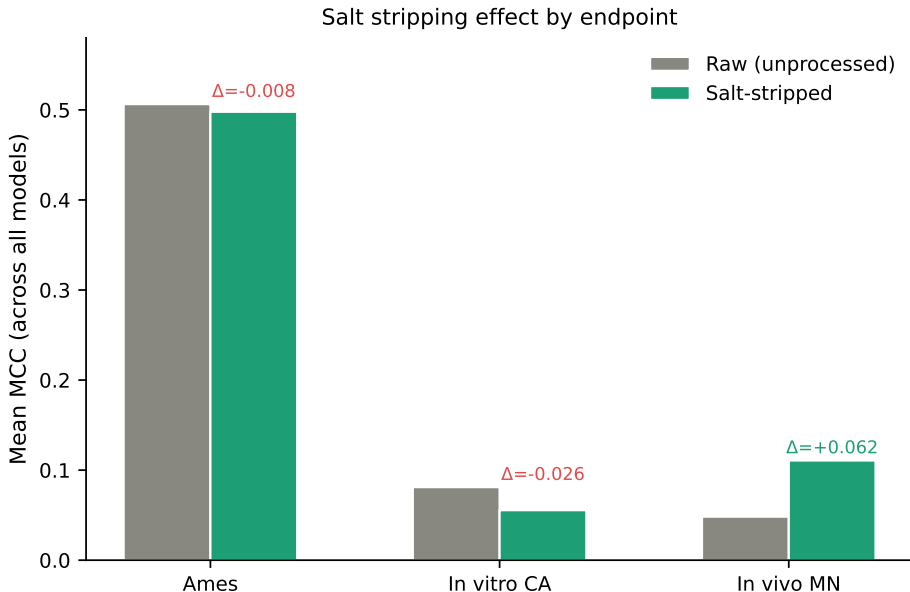


Figure 5: Effect of salt stripping on mean MCC (averaged across all models and feature modes). In vivo improved by +0.063 on average and +0.225 for the best model; Ames mixed; in vitro decreased.

configuration learning curve was flat at  $MCC \approx 0.055$  from 10% to 100% of training data (Figure 7), though this curve was not computed for the salt-stripped configuration.

**Feature importance.** SHAP analysis<sup>41</sup> revealed consistent patterns (Figure 6): physicochemical descriptors dominated the top-5 (alert count and MW for Ames; LogP for in vitro; MW for in vivo), and no individual fingerprint bit appeared in any top-5 list despite fingerprints comprising 50–75% of features. Across all Ames models and scenarios, compact features yielded mean  $MCC = 0.448$  versus 0.523 for FP-2048 ( $\Delta = +0.075$ ), indicating that fingerprints add distributed but individually weak signal.

**Learning curves.** Locked-configuration learning curves (Figure 7) showed three patterns: Ames MCC saturated early (0.543 at 10%, 0.640 at 100%); in vitro showed a noisy upward trend (0.070  $\rightarrow$  0.170); in vivo was flat ( $\approx 0.055$ ).

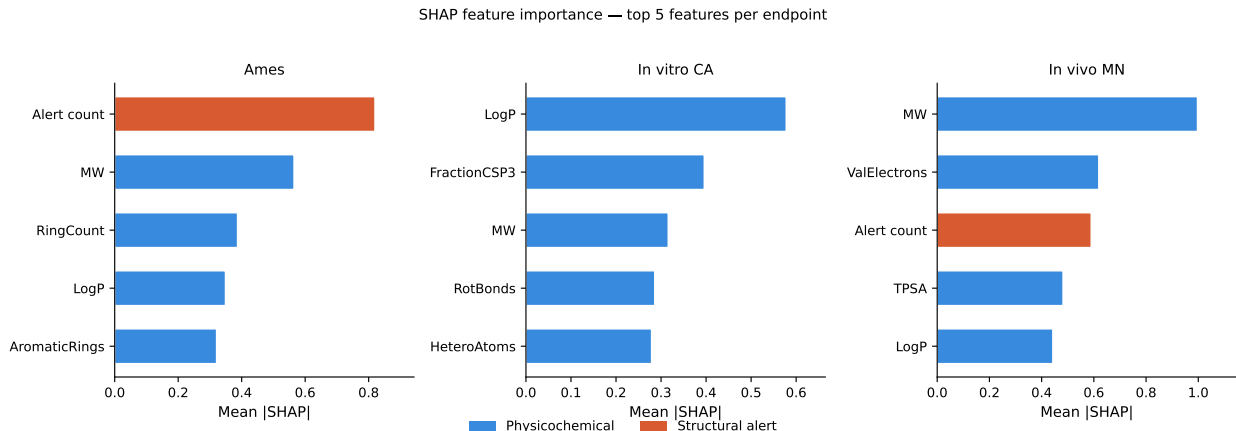


Figure 6: SHAP top-5 features per primary endpoint. Physicochemical descriptors dominate; structural alert count ranks first for Ames. No fingerprint bits appear in any top-5.

## Discussion

### The Performance Variation Hierarchy

The locked-configuration analysis yielded the hierarchy

$$\text{Source heterogeneity} \gg \text{Splitting method} \approx \text{In-domain algorithm choice}$$

with the source-heterogeneity gap ( $\Delta = 0.390$ ) an order of magnitude larger than the split-method gap ( $\Delta = 0.046$ ), itself comparable to the in-domain algorithm spread (MCC range = 0.063). The hierarchy was established on a single Ames benchmark; the LDO  $\Delta$  also reflects asymmetric test sizes and class distributions and should be read as an upper bound; and algorithm robustness was uneven under source shift, so algorithm choice may matter much more in cross-source deployment than the in-domain comparison suggests.

We emphasize that the LDO experiment is not a deployment scenario. In practice, models are trained on all available data regardless of provenance, and the genotoxic potential of a chemical is fundamentally a property of its molecular structure, not of which database it was registered in. The point of cross-source evaluation is rather diagnostic: a large LDO drop reveals that apparent in-domain performance depends on the joint composition of training and test sets rather than on universally learned structure-activity relationships. This mat-

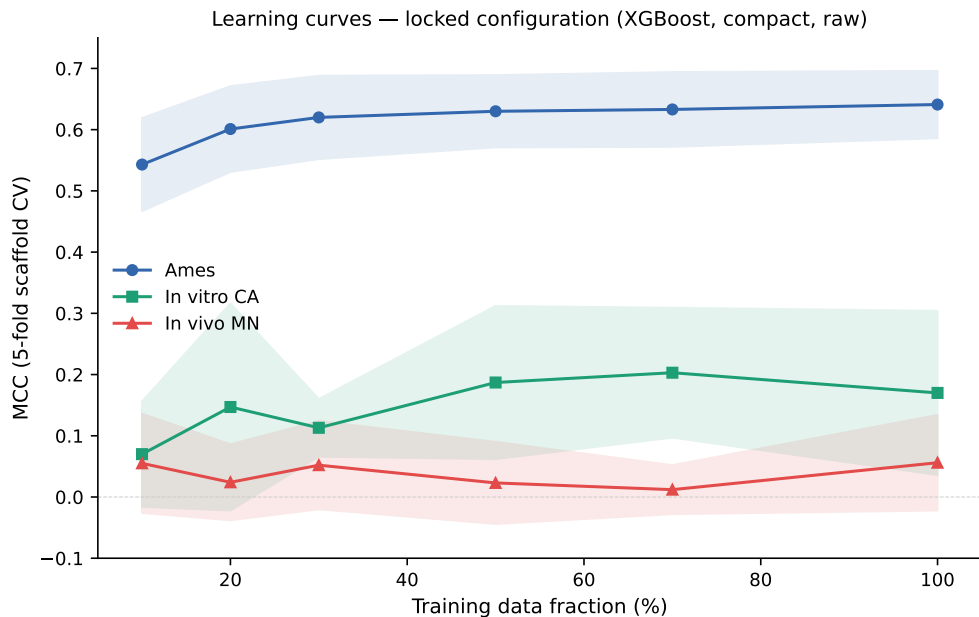


Figure 7: Learning curves (locked configuration). Shaded regions:  $\pm 1$  SD across 5-fold scaffold CV.

ters because new chemicals registered under EU REACH or K-REACH may occupy regions of chemical space that are sparsely represented in pharmaceutical benchmarks, and a benchmark MCC of 0.67 on a mixed-source test set provides little guarantee about the model’s reliability on such compounds. Cross-source LDO acts as a stress test that exposes this dependency and gives a more honest lower bound on out-of-distribution performance.

## What Drives the Source-Heterogeneity Gap?

We identify four non-exclusive hypotheses, in roughly decreasing parsimony. **Prior shift** from the 19.5-fold prevalence gap can mechanically lower MCC by miscalibrating the default decision threshold, consistent with XGBoost reaching sensitivity 0.062 in the industrial→drug direction. **Data quality and labeling heterogeneity** could shift prevalence independently of any chemical difference, since drug-sourced compounds come from curated benchmarks while industrial compounds come from regulatory submissions (ECHA, K-REACH) with varying protocols. **Selection bias** arises because drug candidates are designed for biological

activity while industrial chemicals are selected for stability. **Chemical space difference**<sup>43,44</sup> would mean even a perfectly calibrated cross-source model generalizes poorly.

To estimate the contribution of prior shift, we evaluated each LDO direction at three thresholds: *default* ( $t = 0.5$ ), *prior-corrected* ( $t =$  training positive rate, optimal under pure prior shift), and *oracle* ( $t$  maximizing MCC on the test set, an upper bound on threshold-based recalibration). This used Morgan FP-1024 plus the 13 physicochemical descriptors rather than the full 138-feature compact set, so absolute MCC values differ from Table 5 (notably RF industrial $\rightarrow$ drug: 0.102 here vs. 0.325 in the main analysis), but the within-row three-threshold comparison is internally valid.

The analysis (Table S8) showed that prior-shift correction provided modest improvements (mean +0.075, range  $-0.024$  to  $+0.239$ ); oracle MCC plateaued at 0.27–0.36, well below the in-domain scaffold-CV MCC of 0.624—a residual gap of 0.26–0.35 that no threshold tuning can eliminate. Algorithms differed in default-threshold sensitivity: Random Forest in the industrial $\rightarrow$ drug direction gained +0.243 MCC from default to oracle, while gradient boosting models gained only +0.07 to +0.10. In short, prevalence imbalance accounts for a meaningful but limited portion of the LDO collapse; a substantial residual gap remains. Whichever combination of chemical space, data quality, or selection bias drives the residual, the practical implication is unchanged: published Ames QSAR performance is highly sensitive to source composition, and per-source threshold tuning alone cannot recover in-domain performance under cross-source evaluation.

## In Vivo Micronucleus and Tiered Testing

The dissociation between ranking ability (ROC-AUC = 0.860, PR-AUC = 0.237) and classification reliability (MCC CI crossing zero) means that for regulatory binary hazard classification under ICH M7<sup>3</sup> or REACH,<sup>1</sup> in vivo MN QSAR with 2D descriptors does not achieve sufficient reliability, but the 8.5-fold enrichment over baseline suggests value as one input in tiered screening.<sup>45</sup> The strong preprocessing sensitivity (+0.22 MCC from salt stripping)

means reported in vivo performance depends critically on preprocessing choices that are often undocumented.<sup>20</sup> Mechanistically, micronucleus induction depends on ADME, metabolic activation, tissue distribution, and DNA repair,<sup>46,47</sup> none of which 2D descriptors capture.

## Cross-Endpoint Overlap and Preprocessing

The high compound overlap between endpoints (>90% shared with Ames) implies that performance differences (Ames MCC = 0.555 vs. in vivo 0.311) reflect endpoint predictability from 2D structure rather than differing chemical spaces.<sup>17</sup> Multi-task learning may be a productive direction.<sup>48</sup> Salt stripping was beneficial for in vivo, mixed for Ames, and harmful for in vitro—possibly because some counterions are themselves clastogenically relevant. The QSAR curation literature<sup>19,20</sup> emphasizes standardization, but our results suggest the optimal strategy is endpoint-dependent.

## Practical Recommendations

We propose the following reporting standards for genotoxicity QSAR: (i) scaffold-based splitting; (ii) when source labels exist, cross-source LDO performance with disclosed source composition; (iii) per-endpoint preprocessing optimization; (iv) bootstrap CIs and AD coverage; (v) for class-imbalanced endpoints, both threshold-free (ROC-AUC, PR-AUC) and threshold-dependent (MCC, balanced accuracy) metrics; (vi) testing algorithm superiority claims under domain shift, not just in-domain.

Several constraints bound our findings. We used a single scaffold split rather than repeated holdout, the GCN did not include edge features or pre-training,<sup>49</sup> and we did not evaluate 3D or QM descriptors, metabolite predictions, within-AD vs. outside-AD stratification for LDO, or multiple-comparison correction for the McNemar tests.<sup>23</sup>

## Conclusion

A multi-level evaluation of 375 model configurations across five genotoxicity datasets shows that, for the Ames mutagenicity benchmark studied here, source heterogeneity—the mixing of drug-like and industrial compounds with sharply different prevalence and provenance—drives a substantially larger performance gap ( $\Delta = 0.390$ , scaffold CV  $\rightarrow$  LDO) than the choice between random and scaffold splitting ( $\Delta = 0.046$ ). Threshold-tuning analysis attributes only part of this gap to prior shift, leaving a residual that no recalibration can fix. Algorithm differences are small in-domain (MCC range = 0.063) but become pronounced under source shift, with Random Forest more robust than gradient boosting. In vivo micronucleus models show ranking signal but lack reliable classification utility with current 2D descriptors. We recommend multi-level evaluation with source-awareness as a minimum reporting standard for Ames mutagenicity QSAR.

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# Supporting Information

The Supporting Information is available free of charge. Summary tables are shown below; complete per-configuration data are provided as supplementary CSV files in the code repository.

## Table S0. Endpoint classification criteria

The full mapping of OECD, EPA, and EU Test Guidelines to the three endpoint categories (Ames, in vitro CA, in vivo MN) used for curating all datasets is provided as `table_s0_endpoint_classification.xlsx`. This table was used to assign each raw study record to one of the target endpoints prior to any modeling.

## Table S1. Sampling endpoint results

Table S1: Best model per sampling endpoint (exploratory). Small test sets; interpret with caution.

| Endpoint           | Best model    | MCC [95% CI]      | AUC   | Sens. | $n_{\text{test}}$ |
|--------------------|---------------|-------------------|-------|-------|-------------------|
| In vitro (sampled) | RF / FP-512   | 0.663 [.459,.839] | 0.827 | 0.714 | 69                |
| In vivo (sampled)  | XGB / FP-1024 | 0.832 [.644,1.00] | 0.946 | 0.750 | 47                |

## Table S2. Random versus scaffold split comparison

Table S2: Random vs. scaffold split (XGBoost, compact features, raw SMILES, 5×5 CV).

| Endpoint           | $n$    | Random MCC        | Scaffold MCC      | $\Delta$ |
|--------------------|--------|-------------------|-------------------|----------|
| Ames               | 13,560 | $0.670 \pm 0.010$ | $0.624 \pm 0.049$ | 0.046    |
| In vitro CA        | 1,619  | $0.180 \pm 0.053$ | $0.189 \pm 0.084$ | -0.009   |
| In vivo MN         | 2,153  | $0.181 \pm 0.094$ | $0.104 \pm 0.111$ | 0.077    |
| In vitro (sampled) | 345    | $0.393 \pm 0.085$ | $0.367 \pm 0.085$ | 0.026    |
| In vivo (sampled)  | 231    | $0.525 \pm 0.152$ | $0.429 \pm 0.157$ | 0.096    |

### **Table S3. Full 375-experiment results**

Complete metrics (MCC, bootstrap CI, ROC-AUC, PR-AUC, balanced accuracy, sensitivity, specificity, Brier score, AD coverage) for all 375 configurations are provided in `table_s3_full_results.csv`.

### **Table S4. Hyperparameter tuning comparison**

Default versus tuned MCC for all configurations, including optimized parameter values, are provided in `table_s4_hp_tuning.csv`.

### **Table S5. Applicability domain coverage**

Per-configuration AD coverage (Tanimoto  $\geq 0.3$ , 1024-bit Morgan FP) and mean similarity values are provided in `table_s5_ad_coverage.csv`. Coverage ranged from 48.5% (in vitro) to 58.9% (in vivo) across configurations.

### **Table S6. Cross-endpoint compound overlap**

Pairwise compound overlap and label concordance ( $\kappa$ ) between all endpoint datasets are provided in `table_s6_cross_endpoint.csv`. Overlap with Ames was 97.5% for in vitro and 92.1% for in vivo.

### **Table S7. SHAP feature importance**

SHAP values (mean  $|\text{SHAP}|$ , TreeExplainer) for all features across all tree-based models are provided in `table_s7_shap.csv`.

### **Table S8. LDO threshold-tuning analysis (prior shift vs. source shift)**

To distinguish prior shift (calibration mismatch from differing prevalence) from genuine source shift, we evaluated each LDO direction at three decision thresholds: default ( $t = 0.5$ ), prior-corrected ( $t = \text{training positive rate}$ ), and oracle ( $t$  maximizing MCC on the test

set, providing an upper bound). Per-direction and per-algorithm results are provided in `table_s8_ldo_threshold_analysis.csv`; the analysis script (`ldo_threshold_analysis.py`) is included in the supplementary materials.

*Note on features:* For computational tractability this analysis used Morgan FP-1024 + 13 physicochemical descriptors rather than the full 138-feature compact set used in the main pipeline. Absolute MCC values therefore differ from Table 5 (notably RF industrial→drug: 0.102 here vs. 0.325 in the main analysis). The internal three-threshold comparison within each row is valid because identical features are used across thresholds.

Table S3: LDO threshold-tuning analysis (Morgan FP-1024 + 13 physchem features). Default = MCC at  $t = 0.5$ . Prior-corrected = MCC at  $t$  equal to the training positive rate. Oracle = upper bound MCC across all thresholds. The gap between oracle LDO MCC and the in-domain scaffold-CV MCC ( $\approx 0.624$ ) isolates the irreducible source shift that cannot be corrected by threshold tuning.

| Direction   | Algo.    | MCC default | MCC prior-corr. | MCC oracle | Oracle threshold |
|-------------|----------|-------------|-----------------|------------|------------------|
| Drug → Ind. | XGBoost  | 0.231       | 0.263           | 0.310      | 0.788            |
| Drug → Ind. | LightGBM | 0.241       | 0.264           | 0.304      | 0.781            |
| Drug → Ind. | RF       | 0.250       | 0.299           | 0.359      | 0.690            |
| Ind. → Drug | XGBoost  | 0.200       | 0.267           | 0.273      | 0.041            |
| Ind. → Drug | LightGBM | 0.239       | 0.214           | 0.267      | 0.180            |
| Ind. → Drug | RF       | 0.102       | 0.341           | 0.345      | 0.037            |

Mean improvement (default → prior-corrected): +0.075 (range −0.024 to +0.239). Mean improvement (default → oracle): +0.108 (range +0.028 to +0.243). Oracle MCC range: 0.267–0.359, all substantially below the in-domain scaffold-CV MCC of 0.624.

## TOC Graphic

Some journals require a graphical entry for the Table of Contents. This should be laid out “print ready” so that the sizing of the text is correct. Inside the tocentry environment, the font used is Helvetica 8 pt, as required by *Journal of the American Chemical Society*.

The surrounding frame is 9 cm by 3.5 cm, which is the maximum permitted for *Journal of the American Chemical Society* graphical table of content entries. The box will not resize if the content is too big: instead it will overflow the edge of the box.

This box and the associated title will always be printed on a separate page at the end of the document.